

HouseTab: The Shared Expense Tracker That Actually Settles Up

Summary/Goal of the Application: HouseTab's goal is to solve a financial coordination challenge for people living who are in shared housing. The challenge is that roommates currently rely on informal methods like group chats, mental notes, or basic payment apps to track shared household expenses, splits, and reimbursements over the course of a lease. HouseTab offers a household finance tracker that centralizes shared expenses, tracks running balances between roommates, categorizes spending, and records settlements so that every member of a household has an accurate picture of what they owe and what they're owed.

What is the Problem: Shared housing creates constant financial friction. Roommates regularly cover costs for each other like rent, utilities, groceries, repairs, but rarely have a reliable way to track who owes what over time. General-purpose payment apps like Venmo record individual transfers but offer no running ledger, spending history, or balance tracking across a household. The result is forgotten debts, awkward conversations, and no shared understanding of household finances. HouseTab addresses this by maintaining a persistent, structured record of every expense, split, and settlement throughout the entire lease period.

NOUNS:

Core Entities:

- Household
- User/Roommate
- Expense
- ExpenseSplit
- Category
- Settlement
- Lease
- Recurring/Expense
- Notification
- Tag
- ExpenseTag
- Household Membership

Financial Concepts:

- Balance
- Debt
- Reimbursement
- Running Total

Data & Documents:

- Payment Record
- Spending Summary
- Transaction History
- Monthly Report

VERBS:

- Join — joining a household

- Log — recording a shared expense
- Split — dividing an expense among members
- Settle — resolving a debt between two users
- Track — monitoring running balances
- Categorize — assigning expenses to a category
- View — seeing spending summaries and history
- Add/Remove — managing household membership
- Calculate — computing balances and totals
- Owe — expressing a financial obligation between users

NARRATIVES:

- A User can belong to one Household at a time
- A Household has many Users (roommates)
- A User can log an Expense on behalf of the Household
- An Expense is split among one or more Users through ExpenseSplits
- Each ExpenseSplit records how much a specific User owes for a given Expense
- An Expense belongs to one Category (rent, utilities, groceries, etc.)
- A Settlement records a payment from one User to another to resolve outstanding balances
- The system calculates each User's running balance based on expenses paid and splits owed

Challenge Questions:

- What happens to expenses if a roommate leaves the household mid-lease?
- Can one expense be split unevenly (e.g. one person pays more)?
- How do we handle partial settlements vs. full balance payoffs?

Creative Addition: I extended the system to support uneven expense splits, where roommates can divide costs by custom percentages rather than equal shares. This addresses real-world scenarios like one roommate having a larger bedroom paying more rent, or one person not using a streaming service the others share. In the data model, the ExpenseSplit entity carries a splitAmount attribute that allows each participant's share to be set independently, rather than automatically dividing the total equally.

NOTES

Class Selection:

- User: Roommates who participate in the household and share expenses
- Household: The shared living unit that groups users together
- Expense: A cost paid by one user on behalf of some or all household members
- ExpenseSplit: Resolves the many-to-many between Users and Expenses — records each user's share
- Category: Labels expenses by type for spending analytics
- Settlement: Records a direct payment between two users to resolve accumulated debt

Relationship Decisions:

- Composition: Used for Household -> Expense because expenses cannot exist without a household context
- Association: Used for User -> Settlement and ExpenseSplit -> Expense as these are independent transactional relationships